

Evaluating Privacy Risks of Facial Recognition on Photorealistic Avatars in Social Virtual Reality

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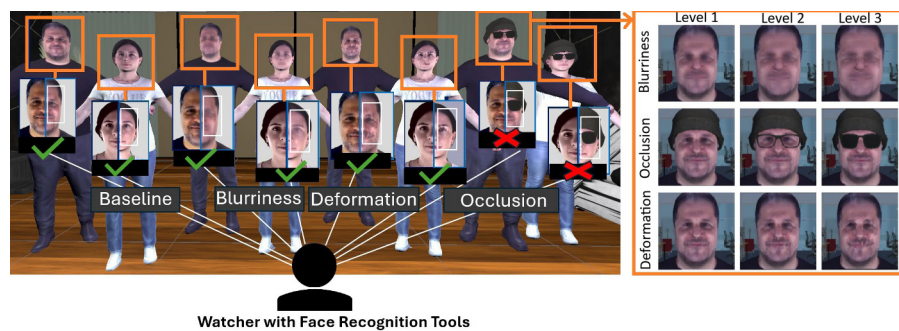


Figure 1: Illustration of the Watcher threat model used in this study; the rightmost panels show three increasing obfuscation levels applied to one example avatar.

ABSTRACT

Photorealistic avatars are increasingly adopted in Social Virtual Reality to enhance presence, yet their close resemblance to users may introduce biometric privacy risks through unwanted face recognition and re-identification. We assess whether state-of-the-art face recognition models can identify avatars derived from real individuals and whether offline visual obfuscation techniques—blurring, mesh deformation, and occlusion—can mitigate this risk. Under the evaluated dataset and threat model, avatars are consistently identifiable without obfuscation, whereas high-intensity facial occlusion substantially reduces identifiability.

Index Terms: Social Virtual Reality, Photorealistic Avatars, Face Recognition, Privacy Risk, Obfuscation Techniques

1 INTRODUCTION

As biometric recognition systems become faster, cheaper, and increasingly embedded in everyday platforms, privacy-enhancing technologies have emerged to reclaim anonymity in public life. A recent example highlights the urgency of this shift: Harvard students demonstrated how Meta’s Ray-Ban smart glasses could be modified into stealth surveillance devices by embedding an open-source facial recognition pipeline, enabling real-time identification of passersby without their consent or awareness.

As physical and virtual worlds converge, biometric vulnerabilities from the physical domain may be mirrored—and potentially amplified—in immersive digital environments [12]. Identity cues that once operated primarily in physical space are increasingly transferred into virtual contexts, where they may persist across sessions and platforms.

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Social Virtual Reality (Social VR) exemplifies this convergence by using realistic, self-similar avatars to enhance social presence, trust, and interaction quality [11]. Prior work has shown that users can be re-identified in VR through behavioral signals such as head, hand, and gait motion, highlighting the privacy risks inherent to immersive systems [9]. However, little attention has been given to visual biometric leakage from photorealistic avatars, as most face recognition and de-identification research focuses on real or stylized faces [5]. Consequently, it remains unclear whether such avatars can be reliably identified by modern face recognition systems or effectively protected through visual obfuscation. This study addresses this gap through a systematic evaluation under a realistic adversarial threat model.

2 MATERIALS AND METHODS

We adopt a Watcher threat model in which an adversary embedded in a Social VR environment visually observes rendered avatars and captures standardized frontal views for automated identity inference using face recognition models (see Figure 1). The adversary has access only to rendered content and no access to raw sensor data or avatar generation pipelines.

We evaluate this threat model in a closed-set identification setting using a probe set of avatar-photo identity pairs and a large distractor gallery. The probe set consists of 10 unique identities (5 male and 5 female, aged 25–45), each represented by a real-world frontal photograph and a corresponding photorealistic avatar. Avatars were generated using a patented high-fidelity multi-view capture-and-reconstruction pipeline. All participants provided written informed consent under institutional ethics approval.

The distractor gallery was derived from CelebA-HQ [7] using automatic offline filtering to retain high-quality frontal adult faces, yielding 3,070 identities. Color and luminance harmonization followed Reinhard et al. [10]. The gallery exhibits typical web-sourced appearance biases, with Individual Typology Angle (ITA)-based skin-tone estimates [3] indicating a skew toward lighter skin

tones.

We evaluated five pretrained face recognition models spanning a range of computational budgets and architectures: MobileFaceNet [1], ArcFace-R50 and ArcFace-R100 [4], MagFace-R100 [8], and TransFace-ViT-L [2]. For each avatar frame, embeddings were computed, and closed-set identification was performed against the full gallery using cosine similarity.

We evaluated three visual obfuscation families—blurring, mesh deformation, and occlusion—each applied at three increasing intensity levels (L1–L3) (see Figure 1). Blurring was applied to facial textures using Gaussian filtering. L1 ($\sigma = 15$, SSIM ≈ 0.70) mildly smooths textures while preserving identity cues; L2 ($\sigma = 30$, SSIM ≈ 0.59) visibly degrades facial detail; L3 ($\sigma = 90$, SSIM ≈ 0.55) strongly obscures key facial features, where SSIM denotes the Structural Similarity Index [6]. Mesh deformation reduces facial geometric detail via mesh simplification (from 3,000 faces). L1 retains $\sim 2,500$ faces with minimal distortion; L2 reduces to $\sim 1,480$ faces with visible contour degradation; L3 collapses to ~ 900 faces, substantially altering facial geometry. Occlusion progressively blocks the upper face using virtual accessories. L1 minimally obstructs the face (hat); L2 partially covers the eyes and forehead (hat + transparent glasses); L3 largely occludes the upper face (hat + dark sunglasses).

Avatars were rendered in Unity under fixed viewing and lighting conditions, with one baseline and nine obfuscated variants per identity. All frames were standardized using face detection, alignment, and resizing prior to recognition analysis.

We report Rank-1 identification accuracy as the primary evaluation metric. All experiments were executed using GPU acceleration on a server with dual Intel Xeon Gold 6130 CPUs, 187 GB RAM, and two NVIDIA Tesla V100 GPUs (32 GB each).

3 RESULTS

Figure 2 summarizes Rank-1 identification accuracy across five face recognition models. Under Baseline conditions, four models achieved 100% accuracy (ArcFace-R50/R100, MagFace-R100, TransFace-ViT-L), while MobileFaceNet achieved 90%, indicating high avatar identifiability.

Obfuscation effectiveness varied by technique, intensity, and model. Blurring (BLR) had limited impact at low-mid levels but reduced accuracy at BLR-3 in a model-dependent manner, whereas mesh deformation (DEF) remained largely ineffective across models. In contrast, occlusion (OCC) caused the strongest and most consistent degradation, with high-intensity occlusion (OCC-3) reducing Rank-1 accuracy to 10–70% depending on the model, indicating that heavy upper-face coverage most reliably limits identifiability.

Per-model Friedman tests confirmed a significant effect of condition on Rank-1 accuracy for all models ($\chi^2(9) = 24.56\text{--}67.17$, $p \leq 0.0035$). However, none of the pairwise Wilcoxon signed-rank tests between the ten experimental conditions (45 pairwise comparisons) remained significant after Bonferroni correction ($\alpha_{\text{adj}} = 0.00111$).

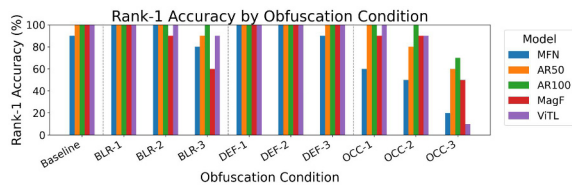


Figure 2: Rank-1 identification accuracy across face recognition models under three obfuscation types—at increasing severity levels (1–3). Models: MobileFaceNet (MFN), ArcFace-R50 (AR50), ArcFace-R100 (AR100), MagFace-R100 (MagF), and TransFace ViT-L (ViTL).

4 DISCUSSION, CONCLUSIONS, AND LIMITATIONS

Under the evaluated conditions, photorealistic avatars can act as biometric proxies, posing privacy risks in Social VR. Obfuscation effectiveness was strongly dependent on the obfuscation technique and its intensity: low- and mid-level blurring and mesh deformation had limited impact, whereas high-intensity facial occlusion consistently produced the largest reductions in Rank-1 accuracy across recognition models. This suggests that obscuring key facial regions may be more effective than degrading texture or geometry alone.

The study is limited by the small number of identities, limited variation in avatar appearance, style, and rendering conditions, standardized frontal views, offline obfuscation, and demographic constraints; therefore, the findings do not generalize to all populations or Social VR scenarios. Nonetheless, this work provides an initial empirical assessment of avatar-mediated identifiability under a realistic adversarial threat model and highlights empirical trade-offs relevant to privacy-aware avatar design.

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